

Intracranial Nanogel Pellets Carrying Temozolomide and Paclitaxel for Adjuvant Brain Cancer Therapy

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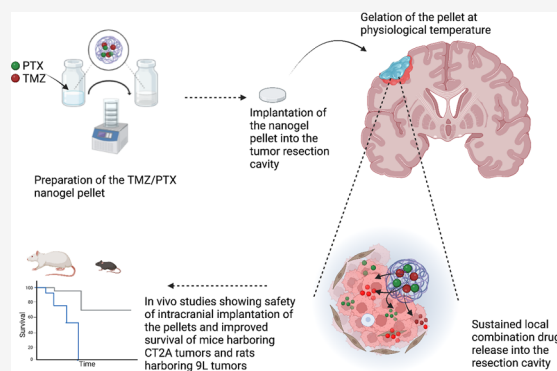
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ABSTRACT: Glioblastoma multiforme is the most frequently diagnosed primary malignant brain tumor. Despite multimodal therapy with surgical resection, radiation therapy, and chemotherapy, recurrence of the tumor is almost always guaranteed due to the infiltrative nature of the disease. Moreover, the blood brain barrier imparts an additional layer of complexity by impeding the delivery of therapeutic agents to the tumor, hence limiting the efficacy of systemically delivered drugs. Hence, to overcome this obstacle and avoid treatment resistance, the local delivery of combination therapies has risen as an appealing adjuvant treatment. The present study describes the creation of a novel PLGA–PEG–PLGA-based nanogel pellet system for the interstitial delivery of Temozolomide (TMZ) and paclitaxel (PTX) to the brain. The nanogel pellet was shown to be stable as a pellet at ambient temperature, absorb water, change to a gel formulation at physiological temperature, and achieve gradual long-term release of TMZ and PTX in vitro. Additionally, in vivo testing of the TMZ/PTX-loaded nanogel pellets in an orthotopic CT2A mouse model and an orthotopic 9L rat model has shown an acceptable safety profile when implanted intracranially and a significant improvement in overall survival.

KEYWORDS: glioblastoma multiforme, glioma, drug delivery, combination therapy, nanogel, thermosensitive gel, animal model



INTRODUCTION

Glioblastoma multiforme (GBM) is the most common primary malignant brain tumor.¹ It accounts for 50.9% of all malignant central nervous system (CNS) tumors.² The current cornerstones of GBM management include maximal safe surgical resection succeeded by chemotherapy in combination with radiation therapy (XRT).³ In certain cases, gross total resection (GTR) of the tumor cannot be achieved, especially if the tumor resides near the eloquent brain areas. Additionally, the infiltrative nature of GBM impedes the eradication of all cancer cells, even with GTR. Hence, XRT and chemotherapy are used as adjuvant treatments once a diagnosis of GBM is established. Current chemotherapy treatments include the implantation of drug-loaded wafers into the resection cavity and/or systemic conventional/targeted therapy.⁴ Nevertheless, despite this multimodal aggressive management, recurrence of the disease is almost always guaranteed.⁵ In fact, the 1 year and 5 year survival rates for patients with GBM are 42.9% and 6.9%, respectively.²

Although Temozolomide (TMZ) remains one of the mainstays of treatment of patients with GBM and presents significant therapeutic efficacy initially, prolonged administration can lead to decreased sensitivity and the development of chemo-resistance.¹ Therefore, combating the resistance of GBM

tumors to TMZ is an important goal in the clinical management of the disease. In this context, paclitaxel (PTX) is a popular chemotherapeutic agent that acts by inhibiting the depolymerization of microtubules within the cell and has shown substantial efficacy in different types of solid malignancies including CNS tumors. Interestingly, Yi et al. have reported that albumin-bound PTX augmented the antitumor effect of TMZ by inducing sustained DNA damage caused by disrupting the expression and nuclear localization of XPC and ERCC1, which are implicated in DNA damage repair.⁶ This indicates that PTX may play a key role in addressing the treatment resistance issues of TMZ. Additionally, Zhao et al. have shown that the addition of TMZ to PTX treatment was able to exert anticancer effects on glioma cells that were initially resistant to the latter through disrupting glucose metabolism.⁷ Yet, a major concern for the

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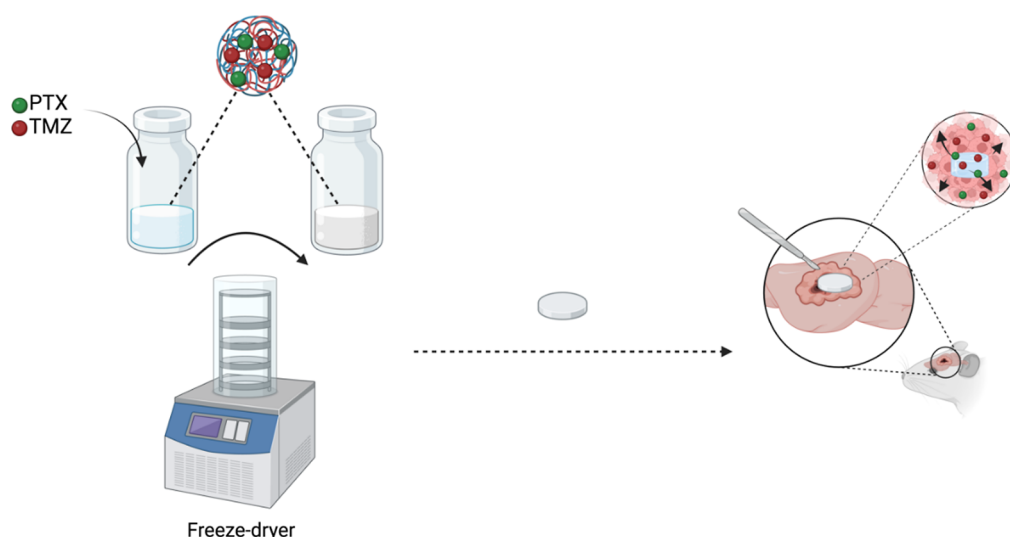


Figure 1. Graphic illustration of the preparation of nanogel pellets carrying TMZ (red circles) and PTX (green circles) followed by implantation of a nanogel pellet in a brain cancer-bearing rodent (illustration generated using BioRender).

delivery of PTX to the brain is its poor penetration of the BBB and the adverse effects associated with its systemic delivery.

The suboptimal delivery of systemically administered chemotherapeutic agents to brain tumors has ignited the search for innovative adjuvant “implantable” therapies that can bypass the BBB altogether.⁸ Gliadel (poly[1,3-bis(carboxyphenoxy)propane-*co*-sebacic acid] (pCPP:SA) polymer with 3.85% carmustine (BCNU)) wafer therapy, which was approved for use in recurrent GBM in 1996 and newly diagnosed high-grade gliomas in 2003 by the Food and Drug Administration, was a pioneering therapy that introduced the local delivery of chemotherapeutic agents into the brain using implantable biodegradable polymers. Despite its efficacy, BCNU wafer therapy has its shortcomings, including its limited capacity to hold a single hydrophilic agent only, uneven distribution of the released drug due to the rigid structure that cannot achieve intimate contact with the cavity boundaries, and inability to maximally cover the resection cavity. Several research efforts expanded on the novel avenue of drug delivery that was opened by Gliadel and investigated different polymers and delivery systems that can achieve a better drug release profile and more convenient handling. One such delivery method is the use of a thermosensitive gel depot (ReGel) that is loaded with PTX, named OncoGel.⁹ Although ReGel, a combination polymer made of poly(lactic-*co*-glycolic acid) (PLGA) and polyethylene glycol (PEG) arranged as a PLGA-PEG-PLGA backbone, is biocompatible and biodegradable and provides a better contact with surrounding tissues in the brain, its use in the operating room poses significant challenges. Specifically, at the operating room’s ambient temperature of 25 °C, ReGel could form a gel in the syringe if no efforts were made to lower the environmental temperature. This could cause syringability issues and inconvenience for administration during a surgical procedure.

Altogether, there is an urgent need to develop a drug delivery system for sustained local chemotherapy in treating malignant gliomas that is (i) therapeutically superior to conventional, systemic chemotherapy, (ii) easy to prepare, (iii) stable for long-term storage, and (iv) convenient for handling in the operating room. In this study, we developed and characterized a nanogel pellet for the local delivery of TMZ and PTX in malignant gliomas and evaluated the treatment effect on survival in animal

models of GBM (Figure 1). The nanogel pellet combines the advantages of wafers and thermosensitive nanogels that codeliver TMZ and PTX. This nanogel pellet was designed to be implanted at the surgical site after tumor resection as an adjuvant therapy. Upon insertion, the pellet is expected to absorb small amounts of physiological fluids, form a gel at 37 °C, and release drugs slowly to eradicate residual cancer cells, prevent recurrence, and extend the survival of treated animals.

MATERIALS AND METHODS

Preparations of Nanogel Pellets Carrying TMZ and PTX. PLGA-PEG-PLGA (1.5k-1.5k-1.5k MW) and PLGA-PEG-PLGA (1k-1k-1k MW) triblock copolymers (Polysciences, West Lafayette, IN) were dissolved in water at 10 and 22 °C, respectively. PLGA-PEG-PLGA (1.5k-1.5k-1.5k MW), PLGA-PEG-PLGA (1k-1k-1k MW), mannitol, and TMZ (Selleckchem, Houston, TX) were mixed in distilled water. PTX (LC Laboratories, Woburn, MA) was solubilized in acetone, transferred into the mixture, and vortexed vigorously for 1–2 min at 20–22 °C. The mixture was then filtered through a 0.2 μ m regenerated cellulose filter, frozen at –80 °C, and freeze-dried (below 0.024 mbar at –52 °C) for 72 h (FreeZone 2.5; Labconco, Kansas City, MO). The lyophilized cake was pressed into pellets, each containing 4 mg of TMZ, 0.2 mg of PTX, 5.4 mg of PLGA-PEG-PLGA (1.5k-1.5k-1.5k MW), 2.7 mg of PLGA-PEG-PLGA (1k-1k-1k MW), and 1.7 mg of mannitol. For the *in vivo* study on a mouse model, each mini pellet weighed 5 mg and carried 1.4 mg of TMZ and 0.07 mg of PTX. The size of the pellets was measured by a digital caliper (NEIKO stainless hardened).

The content of drugs was analyzed by reverse phase high-performance liquid chromatography (RP-HPLC) using an Agilent 1220 Infinity LC system (Shimadzu, Japan) following previously established method.¹⁰ Briefly, 5 μ L of each sample was introduced into an Eclipse C18 column (3.0 $\times$ 150 mm, 3.5 μ m, Agilent, Santa Clara, CA). For this procedure, the flow rate was set at 0.8 mL/min and the column temperature was set at 25 °C. A gradient mode was utilized to perform the separation of PTX and TMZ: mobile phase A was constituted of 0.1% trifluoroacetic acid (TFA; Sigma-Aldrich, St. Louis, MO) in acetonitrile (Fisher Scientific, Hampton, NH), while mobile

phase B was constituted of 0.1% TFA in HPLC grade water. The initial gradient was made of 90% mobile phase A and was maintained for 5 min, then lowered to 10% over the span of 5 min, and kept at 10% for a period of 3 min. Finally, the gradient was increased to 90% mobile phase A during the last 5 min to complete the protocol. TMZ and PTX were detected at 260 nm and eluted at 0.8 and 15 min, respectively.

Characterizations of Nanogel Pellets and Sol–Gels Carrying TMZ and PTX Pellet-To-Gel Transition. The transition of nanogel pellets from a solid dosage form to a gel was monitored by storing a pellet (14 mg/unit) in 150 μ L of artificial cerebrospinal fluid (CSF, pH 7.2) at 22 and 37 °C for 5 min. The ratio of the weight of CSF absorbed by pellets to the weight of the pellet was calculated.

Gelation Temperature and Viscosity Measurement. The lyophilized cake containing 4 mg of TMZ and/or 0.2 mg of PTX with 6% w/v triblock copolymers was reconstituted with 1 mL of distilled water. The reconstituted sol–gel mixture was transferred to a glass scintillation vial and monitored over an increasing range of temperatures to detect the temperature at which gel formation occurred. In specific, gel formation was defined as the point after which the sample would not drop or slide along the vial's wall upon inversion. This procedure was performed over a hot plate, and the plate's temperature was varied between 22 and 40 °C. The scintillation vials were placed in a water-filled glass dish over the plate, and the temperature of the water was monitored using a probe throughout the procedure.

To measure the sample's torque at various rotational speeds, a cone rheometer (DVNext Wells-Brookfield Cone Rheometer; AMETEK Brookfield Middleboro, MA) was used. This torque meter has the capability to detect the resistance to the rotation of a cone over a stationary plate that is imparted by the placement of the fluid sample between the two. This allows inference of the viscosity profile of the sample tested. To perform this procedure, 1 mL of sample fluid was used, and the rotational speed of the cone was set to 10 rpm. Each sample was run at different temperatures (10, 22, and 37 °C), and the respective readings were made after 5 min.

Particle Size. Dynamic light scattering (DLS) was used to measure the Z-average diameters of the synthesized sol–gels at different temperatures (10, 25, and 37 °C) using a ZetaSizer Nano ZS (Malvern Instruments, United Kingdom) at a 173° detection angle and using a He–Ne ion laser incident beam (4 mW, λ_{max} = 633 nm). This system employs an avalanche photodiode detector that can capture the scattering intensity and is subsequently analyzed using a digital autocorrelator. The analysis is performed based on the Stokes–Einstein equation to produce a correlation function. The samples used for this process were previously diluted, aiming for a final polymer concentration of 12%.

Drug Release Profile. To estimate drug release profiles, a pellet (5 mg) or a sol–gel (25 μ L) carrying 1.43 mg of TMZ and 0.07 mg of PTX was placed in 2 mL of artificial cerebrospinal fluid (pH 7.2) at 37 °C with gentle shaking. At various time points, 0, 10 min and 1, 2, 5, 8, 24, 48, 72, and 96 h, 10 μ L of buffer was collected (replenished with fresh buffer), and the amount of drug released was quantified by HPLC analysis.

In Vitro Cell Cytotoxicity of TMZ, PTX, and TMZ/PTX. In vitro cell cytotoxicity of TMZ, PTX, and both drugs combined was assessed in 9L gliosarcoma (organism: rat, cell type: glial cell, tissue: brain, morphology: fibroblast) and a patient-derived glioblastoma [GBM patient-derived xenograft (PDX) 43] cells

(Jann Sarkaria Translational Neuro-Oncology Laboratory, Mayo Clinic, Rochester, MN). GBM PDX 43 cells were identified as primary glioblastoma (Grade IV, Fibrillary) cells extracted from a white male patient (right temporal) who never received treatment. Rat 9L gliosarcoma cells (Brain Tumor Research Center at the University of California, San Francisco) were cultured in Dulbecco's modified Eagle's medium supplemented with 4500 mg/L glucose, 4 mM L-glutamine, 110 mg/L sodium pyruvate, 10% fetal bovine serum, and 1% penicillin/streptomycin/amphotericin B solution. PDX 43 was cultured in Eagle's Minimal Essential Medium with L-glutamine supplemented with 4500 mg/L glucose, 4 mM L-glutamine, 110 mg/L sodium pyruvate, 10% fetal bovine serum, and 1% penicillin/streptomycin/amphotericin B solution. Cells were cultured in a humidified incubator at 37 °C under an atmosphere of 5% CO₂. Once cells reached >90% confluency, they were harvested after trypsinization. Subsequently, 9L (5×10^3 cells/well) and PDX 43 (2×10^3 cells/well) were seeded in 96-well plates. 24 h after seeding, the cells were treated with TMZ, PTX, and a combination of TMZ/PTX at final concentrations of 1, 10, 100, 1,000, and 10,000 nM. After 48 h, cell viability was determined using CellTiter-Blue Reagent (Promega, Madison, WI) as per the manufacturer's instructions. After 3h of incubation, fluorescence was subsequently measured at 560/590 nm using a Spark 10 M TECAN plate reader (Tecan Group Ltd., Männedorf, Switzerland). The CompuSyn software (Composyn Inc., Paramus, NJ) was used to calculate the half-maximal inhibitory concentration (IC₅₀). The same software was also used to apply the Chou–Talalay method, which allows for the calculation of a combination index (CI) for drug combinations to characterize their interactions. In specific, CI < 1 indicates synergism, CI = 1 indicates an additive effect, and CI > 1 indicates a subadditive effect.

In Vivo Safety Study. Female Fischer 344 rats were purchased from Charles River Laboratories (Frederick, Maryland). At the beginning of every study, the rats weighed 140–200 g each. Rats were put under anesthesia using a 0.45 mL intraperitoneal injection of a solution composed of ketamine HCl (75 mg/kg), xylazine (7.5 mg/kg), and ethanol (14.25%) in a sterile 0.9% NaCl solution. The animals were also given a subcutaneous (SC) injection of 1–1.2 mg/kg buprenorphine SR. Upon confirmation of proper anesthesia, the surgical site was shaved and the scalp was prepped using a Prepodyne solution. Subsequently, the scalp was incised at the midline, and the galea was displaced laterally to expose the left parietal bone. A burr hole with a 3 mm diameter was drilled in the left parietal bone using a TPS universal Drill (Stryker Corp); the center of the hole was located 3 mm lateral to the sagittal suture and 5 mm posterior to the coronal suture. Careful drilling was performed under the guidance of a Zeiss surgical microscope (Zeiss OPMI 6-CFC surgical microscope) until the dura was visualized. Following durotomy, 1, 2, or 3 nanogel pellets carrying TMZ and PTX at 4 and 0.2 mg, respectively, were inserted, and surgical staples were then used to close the scalp incision. After the procedure, the rats were monitored daily for deleterious signs according to the Johns Hopkins Animal Care and Use Guidelines. Once these signs were noticed, the animal was euthanized according to the protocol, and the brain and major organs were harvested and preserved in 10% formalin. Animals were weighed and checked for neurological deficits or behavioral changes three times per week.

In Vivo Longevity of the Nanogel Pellets. To test the longevity of the pellets when implanted intracranially, we

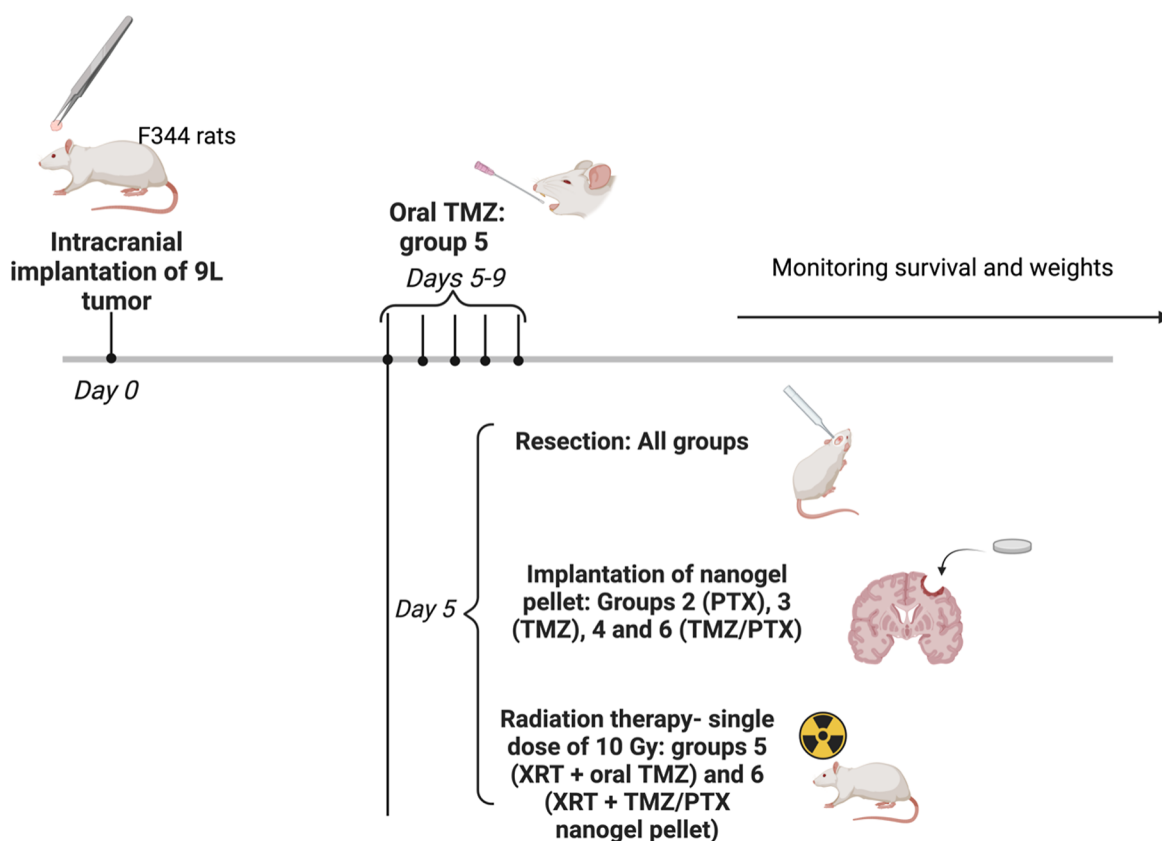


Figure 2. Schematic illustration of the in vivo efficacy study protocol for nanogel pellets on long-term survival in a rodent brain tumor model (illustration generated using BioRender).

prepared pellets that are tagged with a fluorescent dye (DiR, 8 μ g DiR per pellet). Ten female C57BL/6, 7 week-old mice, weighing 20–25 g, purchased from Charles River Laboratories, were used. Mice were anesthetized with a 0.2 mL intraperitoneal injection of a solution composed of ketamine HCl (25 mg/kg), xylazine (2.5 mg/kg), and ethanol (14.25%) in a sterile 0.9% NaCl solution. The animals were subsequently shaved, and the surgical site was prepped with Prepodyne. They received an analgesic dose of buprenorphine SR, 0.5–1.0 mg/kg, given subcutaneously. After that, a midline scalp incision was made and the galea was displaced laterally to visualize the left cranium. A 2 mm burr-hole was drilled in the left parietal bone with its center 2 mm lateral and 2 mm posterior to bregma. With the aid of a Zeiss surgical microscope (Zeiss OPMI 6-CFC surgical microscope), a durotomy was performed and the nanogel pellets were implanted in the brain parenchyma. Finally, the skin was closed with one sterile autoclip. Subsequently, on prespecified time points (day 0, 5, 10, 20, and 30), 2 mice were randomly selected, euthanized, and perfused with chilled phosphate buffered saline, and their brains were harvested. The harvested brains were imaged ex vivo to detect DiR fluorescence (excitation: 754 nm; emission: 778 nm) using the IVIS Lumina Series III system.

In Vivo Efficacy of Nanogel Pellets with or without XRT in a Rodent Brain Tumor Model. The 9L gliosarcoma tumor was propagated as a solid tumor in the flank of the donor F344 rats. For intracranial implantation, the donor rat was euthanized, and the tumor was excised using a sterile technique. The 9L tumor was cut into 2 mm³ pieces and preserved at 4 °C during the procedure. For intracranial implantation, the intracranial surgical procedure as described above was followed. Once the

burr hole was drilled, a small piece of the cortical and subcortical regions was aspirated. After hemostasis was achieved, the tumor piece was placed into the defect. The incision was then closed with surgical clips, and the animal was observed.

Five days after tumor implantation, the scalp incision was reopened, and the tumor tissue was resected using a sterile glass pipet and gentle suction. Rats were randomly assigned into four treatment groups and treated as follows: (i) control (resection only) ($n = 8$), (ii) resection + intracranial 3.85% BCNU: pCPP:SA wafer (10 mg) ($n = 8$), (iii) resection + intracranial TMZ (4 mg) nanogel pellet ($n = 9$), and (iv) resection + intracranial TMZ (4 mg)/PTX (0.2 mg) nanogel pellet ($n = 10$). The animals were observed daily and weighed twice a week for 120 days (intracranial implantation of 9L considered day 0). Animals alive on day 120 were considered long-term survivors.

In a separate study (Figure 2), rats intracranially implanted with the 9L tumor were randomized into 6 groups and treated as follows: (i) control (resection only) ($n = 7$), (ii) resection + intracranial PTX (0.2 mg) nanogel pellet ($n = 9$), (iii) resection + intracranial TMZ (4 mg) nanogel pellet ($n = 8$), (iv) resection + intracranial TMZ (4 mg)/PTX (0.2 mg) nanogel pellet ($n = 6$), (v) resection + XRT + oral TMZ ($n = 7$), and (vi) resection + XRT + intracranial TMZ (4 mg)/PTX (0.2 mg) nanogel pellet ($n = 8$). For the XRT procedure, rats were anesthetized as described previously, positioned individually into an XSTRAHL Small Animal Radiation Research Platform (SARRP) machine, and exposed to a CT-guided single radiation dose of 10 Gy on the day of surgical resection. A 10 $\times$ 10 mm X-ray beam was used, and it was targeted 5 mm inferior to the drilled hole in the skull. The energy employed for the treatment beam was 220 kVp and 13 mA using a 0.5 mm Cu filter. Oral TMZ was



Figure 3. Various nanogel formulations: (a) lyophilized cake, (b) rehydrated sol at 22 °C, (c) nanogel pellets, and (d) gel formed from a nanogel pellet at 37 °C.

administered using a bulb-tipped oral gavage needle at a dose of 10 mg/kg daily for 5 days, starting the day of resection.

In Vivo Efficacy of Nanogel Pellets on Long-Term Survival in a Murine Brain Tumor Model. Female C57BL/6 7 week-old mice, weighing 20–25 g, purchased from Charles River Laboratories, were used for all studies. Mice were anesthetized, prepped, and incised as detailed previously. A 2 mm burr-hole was drilled in the left parietal bone with its center 2 mm lateral and 2 mm posterior to bregma. The animals were then placed on a stereotactic frame, and using a 26G Hamilton syringe, 3 μ L of 100,000 firefly luciferase-tagged CT2A murine glioma cells was injected at a depth of 2.5 mm over a period of 1 min. The incision was closed by using a sterile autoclip.

Tumor implantation was confirmed using bioluminescence-based in vivo imaging. Specifically, 5 days after the CT2A injection, the mice received an intraperitoneal injection of 0.2 mL of 15 mg/mL D-luciferin solution each and then were imaged using the PerkinElmer Lumina series III In Vivo Imaging System (5 min after injection). The region of interest was limited to the head, and total bioluminescence per mouse was measured. Animals with tumors with a total flux $>10^7$ were stratified into 4 treatment groups, ensuring similar averages and standard deviations between the groups. Mice were then anesthetized as described previously, the scalp incision was reopened, and mice were treated as follows: (i) control (sham surgery) ($n = 9$), (ii) intratumoral TMZ (1.4 mg) nanogel pellet ($n = 8$), (iii) intratumoral PTX (0.07 mg) nanogel pellet ($n = 8$), and (iv) intratumoral TMZ (1.4 mg)/PTX (0.07 mg) nanogel pellet ($n = 8$). Animals were observed daily and weighed two times per week. The tumor burden was assessed by measuring bioluminescence, as described previously, three times per week. If animals showed deleterious signs, they were euthanized and their brains were harvested and preserved in 10% formalin.

Statistical Analysis. Survival was plotted using Kaplan–Meier curves using GraphPad Prism (ver. 9) and analyzed using the Log-Rank Mantel–Cox test in IBM SPSS (version 29.0). For analysis of a continuous variable among 3 or more groups, the one-way ANOVA test was used followed by a Fisher’s LSD test without correction for multiple comparisons. Comparison of two normally distributed continuous variables was conducted using an unpaired *t*-test along with a Levene’s test for the equality of variances. Comparison of a categorical variable among two groups was performed using Fisher’s exact test. Significance was defined as a *p*-value less than 0.05.

RESULTS

Characterizations of Nanogel Pellets and Nanogels (Sol–Gels) Carrying TMZ and PTX. Nanogel pellets carrying TMZ and PTX were prepared by pressing the lyophilized cake of nanogels into pellets (Figure 3). For the 14 g pellets, each pellet weighed 14 ± 1 g and measured 2.71 ± 0.13 mm in diameter and

1.02 ± 0.12 mm in thickness. Based on visual observation, regardless of the payloads, the sol–gel form (rehydrated sol) of the formulation gained the viscosity and formed a gel at >32 °C but remained as a sol at ambient temperature (Figure 3; Table 1). Nanogels diluted under a gelation concentration, below 20%,

Table 1. Measurements of Gelation Temperature, Viscosity at 10, 22, and 37 °C, and Particle Size at 10, 22, and 37 °C for TMZ/PTX, TMZ, PTX, and Empty Nanogels

formulation	TMZ/PTX nanogel	TMZ nanogel	PTX nanogel	empty nanogel
gelation temperature (°C)	32.8	33.9	33.2	36.3
viscosity at 10 °C (cP)	3.92	8.50	35.32	9.16
viscosity at 22 °C (cP)	30.74	21.58	88.29	23.54
viscosity at 37 °C (cP)	96.79	119.00	133.00	98.10
particle size at 10 °C (nm) (PDI)	236.0 (0.619)	284.1 (0.390)	372.9 (0.117)	53.6 (0.463)
particle size at 22 °C (nm) (PDI)	242.1 (0.895)	222.4 (0.994)	1020 (0.620)	143.1 (0.232)
particle size at 37 °C (nm) (PDI)	131.7 (0.195)	128.0 (0.276)	145 (0.219)	87.4 (0.406)

formed nanoparticles: The particle sizes of diluted nanogels (12%) at 37 °C were ca. 132 nm for TMZ/PTX nanogels, ca. 128 nm for PTX nanogels, and ca. 87 nm for empty nanogels. All nanogels gained viscosity when the temperature rose from 10 to 37 °C. The viscosity of nanogels carrying TMZ and PTX was ca. 3.92 cP at 10 °C, 30.74 cP at 22 °C, and 96.79 cP at 37 °C, showing the most significant increase (a 25-fold increase) in its viscosity at 37 °C. TMZ nanogels, PTX nanogels, and empty nanogels showed the viscosity increased at 14-, 3-, and 11-fold, respectively, at 37 °C. The gelation temperature of TMZ/PTX nanogels was the lowest (32.8 °C), and that of empty nanogel was found to be the highest (36.3 °C) (Table 1).

To observe the visual transition from a solid dosage form to a gel, the nanogel pellets (14 mg/unit) were stored in 150 μ L of artificial cerebrospinal fluid (pH 7.2) at 22 and 37 °C for 5 min (Figure 4). At 22 °C, the pellet did not form a gel but dissolved slowly (visual dissolution rate: empty pellet > TMZ pellet > TMZ/PTX pellet), disintegrating from the bottom of the pellet. At 37 °C, the pellet absorbed water, formed a gel, and decreased the volume of the pellet slowly over the 5 min period (visual dissolution rate: TMZ pellet > empty pellet > TMZ/PTX pellet). The pellet remaining in the fluid at 37 °C after 5 min showed gel-like integrity (viscous and tacky). The ratios of the weight of CSF absorbed by the pellet over the weight of the pellet were 1.4 (e.g., 18.2 mg/13 mg) for empty pellet, 1.8 for TMZ/PTX pellet, and 1.6 for TMZ pellet. The Supporting Information (video clips showing nanogel pellets stored in artificial cerebrospinal fluids at 22 and 37 °C) is available online.

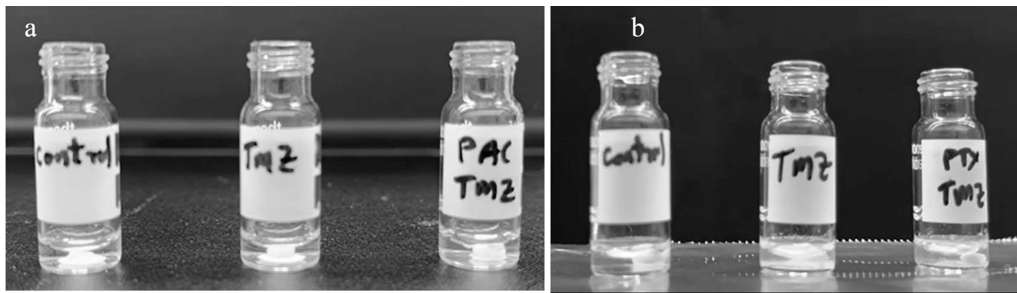


Figure 4. Nanogel pellets at 22 °C (a) and 37 °C (b) stored in artificial cerebrospinal fluid for 5 min (Supporting Information).

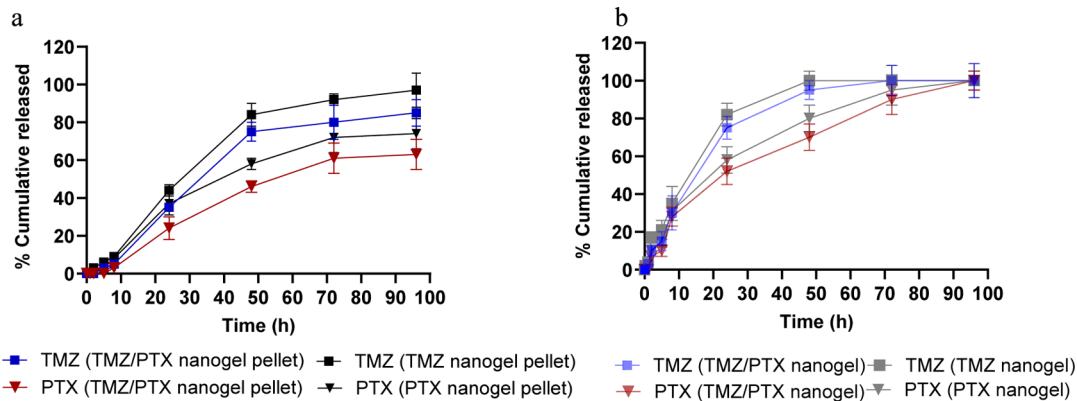


Figure 5. Drug release profiles for TMZ and PTX from nanogel pellets (a) and nanogels (sol–gels) (b).

Drug release kinetics were plotted for various formulations, and statistical analysis results using paired *t*-test are shown as follows (Figure 5): TMZ from TMZ nanogel pellet vs TMZ from TMZ/PTX nanogel pellet *p* value = 0.082, PTX from PTX nanogel pellet vs PTX from TMZ/PTX nanogel pellet *p* value = 0.0109, TMZ from TMZ/PTX nanogel pellet vs PTX from TMZ/PTX nanogel pellet *p* value = 0.0349, TMZ from TMZ sol–gel vs TMZ from TMZ/PTX sol–gel *p* value = 0.0067, PTX from PTX sol–gel vs PTX from TMZ/PTX sol–gel *p* value = 0.0084, and TMZ from TMZ/PTX sol–gel vs PTX from TMZ/PTX sol–gel *p* value = 0.0408. Drug release profiles were, in general, the best fit in a first-order mathematical model (Table 2). The half-time, which is the time taken to release 50% of the

Table 2. Half-Time (h) for Drug Release and the Correlation Coefficient on a First-Order Release Model

formulations	half-time (h)	R^2
TMZ (TMZ/PTX pellet)	32.8	0.9636
PTX (TMZ/PTX pellet)	60.4	0.9752
TMZ (TMZ pellet)	21.3	0.9930
PTX (PTX pellet)	44.0	0.9724
TMZ (TMZ/PTX sol–gel)	12.1	0.9671
PTX (TMZ/PTX sol–gel)	20.2	0.9272
TMZ (TMZ sol–gel)	8.5	0.8846
PTX (PTX sol–gel)	17.3	0.9797

payload, for PTX from TMZ/PTX nanogel pellets was ca. 60.4 h. The half-time for PTX from TMZ/PTX nanogels was ca. 20.2 h, approximately 3-fold shorter compared to that for PTX from TMZ/PTX nanogel pellets. A similar pattern was observed for TMZ from TMZ/PTX nanogel pellets (32.8 h) vs TMZ from TMZ/PTX nanogels (12.1 h). Comparing TMZ from TMZ/PTX pellets to TMZ from TMZ pellets, the half-time was

delayed for the TMZ/PTX pellet (32.8 h vs 21.3 h). A similar pattern was observed for PTX from TMZ/PTX pellets vs PTX from PTX nanogel pellets (60.4 h vs 44.0 h). This indicates that the release kinetic profile of TMZ is more rapid than that of PTX. The release kinetic profiles of both payloads were more rapid from nanogels than those from nanogel pellets.

The in vitro cell cytotoxicity results on 9L rat gliosarcoma and GBM PDX 43 demonstrated that TMZ alone was not effective in eradicating cancer cells (Table 3). However, when combined

Table 3. In Vitro Cell Cytotoxicity of Free Drugs on the 9L Gliosarcoma and PDX 43 GBM Cells

9L Gliosarcoma	TMZ + PTX	TMZ	PTX
IC ₅₀ (nM)	5.42	1441.48	989.49
combination index	0.00377		
GBM PDX 43	TMZ + PTX	TMZ	PTX
IC ₅₀ (nM)	22.77	141000	0.42676
combination index	0.59961		

with PTX at a 4:0.2 w/w ratio, it showed a significant synergistic effect on both cells (PDI < 1). The synergism appeared to be more significant on 9L than on GBM PDX 43.

In Vivo Safety, Longevity, and Efficacy Results. The safety of the TMZ/PTX-loaded nanogel pellets was evaluated by the intracranial implantation of a variable number of pellets (1 vs 2 vs 3 pellets) into naive rats that had no tumors (Figure 6). The weights of the rats were monitored for 30 days postimplantation. All groups showed a consistent increase in average weights over this period (Figure 6). The in vivo implantation of fluorescently labeled nanogel pellets shows that the delivery system achieves gradual biodegradation intracranially and has substantial longevity lasting for at least 30 days in the mouse brain (Figure 7). Additionally, in all of the following experiments involving the

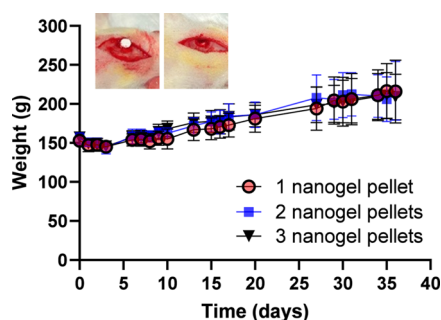


Figure 6. Body weight changes of F344 rats after intracranial implantation of 1-, 2-, or 3-nanogel pellets carrying TMZ and PTX; inset depicts photographs of intracranial implantation and insertion of the pellet.

treatment of mice and rats implanted with brain tumors, none of the treated groups showed a significant decrease in the average weight of the animals (Figures 8–10).

To evaluate the *in vivo* efficacy of the drug-loaded nanogel pellets, 80 F344 rats (35 in the first study and 45 in the second study) were implanted with 9L gliosarcoma tumors and 33 C57Bl6 mice were implanted with the aggressive murine glioma CT2A cell line. The first rat study (Figure 8) showed that all three treatment groups had a significantly better overall survival than the control group which received resection only (using log-rank test, BCNU wafer vs control p -value = 0.002; TMZ nanogel pellets vs control p -value = 0.0005; TMZ/PTX nanogel pellets vs control p -value = 0.003) (Figure 8). There were no significant differences in overall survival among the treatment groups. Although all control rats were dead by day 25 postimplantation, the BCNU wafer, TMZ nanogel, and TMZ/PTX nanogel groups had 37.5%, 11.1%, and 40% long-term survivors (surviving >120 days), respectively.

The second rat study (Figure 9) showed a significant survival advantage for TMZ nanogel pellets, PTX nanogel pellets, oral TMZ + XRT, and TMZ/PTX nanogel pellets + XRT when added to resection as compared to resection alone (using log-rank test, PTX nanogel pellets vs control p -value = 0.046, TMZ nanogel pellets vs control p -value = 0.005, oral TMZ + radiation vs control p -value = 0.010, TMZ/PTX nanogel pellets + radiation vs control p -value = 0.012) (Figure 9). There were no

significant differences in overall survival among the treatment groups. Interestingly, when it comes to long-term survival (>120 days), PTX nanogel pellets, TMZ/PTX nanogel pellets, and TMZ/PTX nanogel pellets + XRT groups had 22.2%, 33.3%, and 37.5% long-term survival, respectively, while control and oral TMZ + XRT groups did not survive past days 35 and 46 postimplantation, respectively (Figure 9).

As for the mouse study, to confirm tumor implantation and ensure equal distribution of tumor burdens among the groups, bioluminescence of the firefly luciferase-tagged CT2A cells was measured. There was no statistically significant difference in the average bioluminescence among the four groups on the day of treatment (ANOVA p -value = 0.910). The group that received the combination TMZ/PTX nanogel pellet had a significantly enhanced overall survival compared to the control group that received sham surgery (p -value = 0.037). Additionally, the combination TMZ/PTX nanogel pellet group had a significantly prolonged median survival of 17 days [95% confidence interval (CI): 14.6–19.4] as compared to the control (12 days [95% CI: 11.3–12.7]), TMZ nanogel pellets (12 days [95% CI: 10.2–13.8]), and PTX nanogel pellets (12 days [95% CI: 10.7–13.3]) (Figure 10). Bioluminescence was measured to assess the tumor burden three times per week throughout the study. Although all groups started with a similar average bioluminescence, the control group had the steepest increase in average bioluminescence over time (Figure 10); however, this difference did not reach statistical significance.

DISCUSSION

A hallmark of GBM's growth is its aggressive and invasive behavior and resistance to treatment. Even with maximal surgical resection, radiation therapy, and chemotherapy, recurrence is inevitable. An additional obstacle that faces the effective treatment of this tumor is the blood brain barrier. The blood brain barrier hinders the safe delivery of therapeutic agents to the tumor. The interstitial delivery of carmustine using a biodegradable pCPP:SA polymer (Gliadel) changed the field of local sustained drug delivery in the setting of glioblastoma and opened the door for further investigation into the realm of implantable depots for interstitial drug delivery into the brain.^{11–14} To further expand on the potential of local drug delivery, a combination of different drug candidates has been

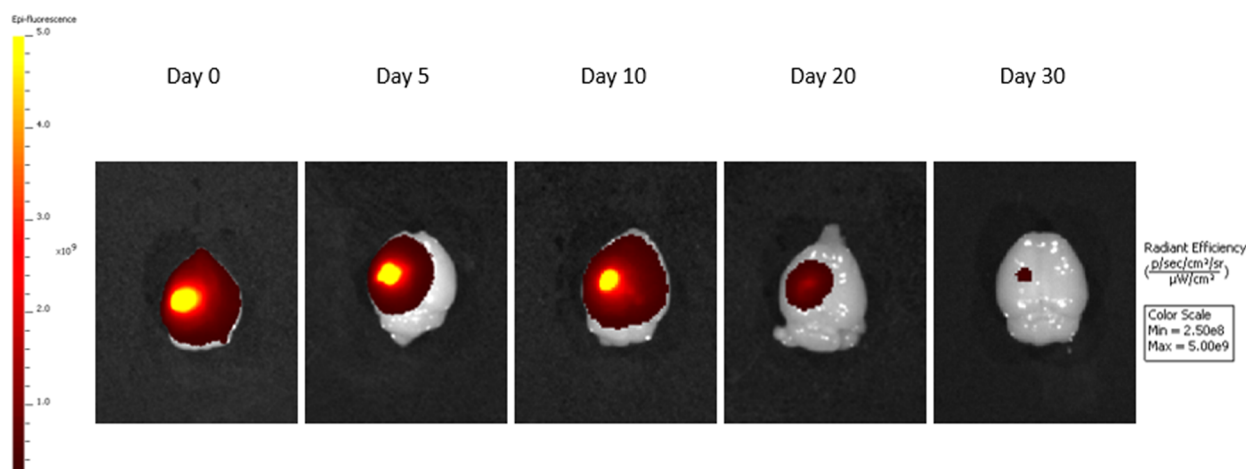


Figure 7. Change in ex vivo fluorescence emitted from the brain harvested on days 0, 5, 10, 20, and 30 after the intraparenchymal implantation of nanogel pellets tagged with DiR. The minimum and maximum radiance efficiency values were set consistently for all images at 2.5×10^8 and 5.0×10^9 , respectively.

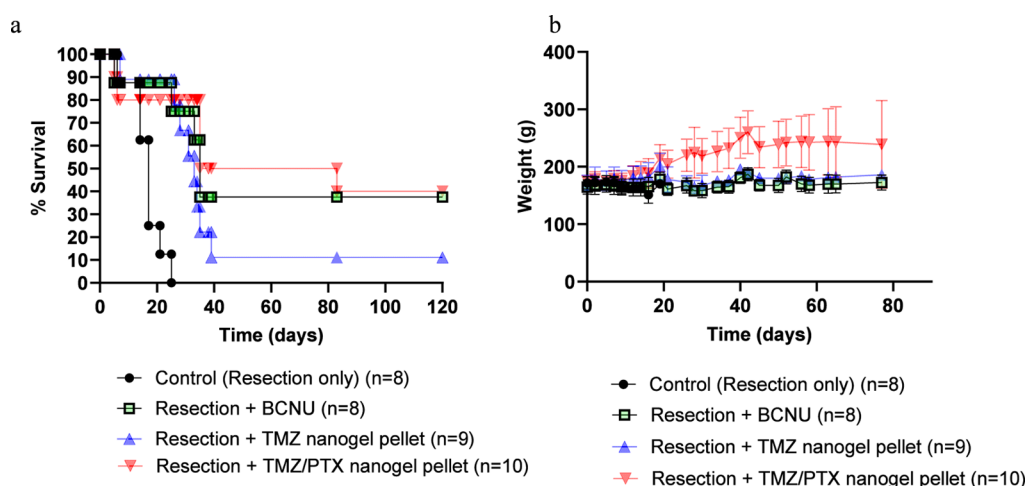


Figure 8. First study of the in vivo efficacy of the drug-loaded nanogel pellets in F344 rats implanted with 9L gliosarcoma tumors: (a) Kaplan–Meier plot for long-term survival and (b) graph showing the variation of weight measurements of animals over time.

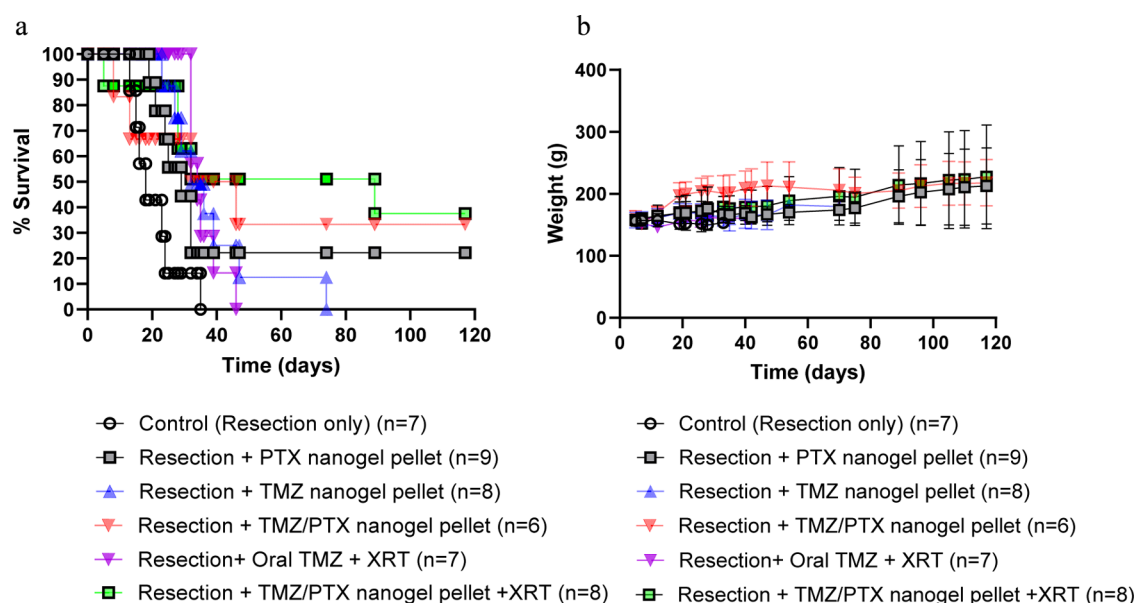


Figure 9. Second study of the in vivo efficacy study of the drug-loaded nanogel pellets in F344 rats implanted with 9L gliosarcoma tumors: (a) Kaplan–Meier plot for long-term survival and (b) graph showing the variation of weight measurements of animals over time.

widely explored and has shown promising results.^{15–18} Similarly, in this study, the combined local delivery of TMZ and PTX via the nanogel pellet formulation offers a significantly better median survival than the local delivery of either agent alone in mice implanted with CT2A tumors. An important characteristic of this combination is that the two agents have different mechanisms of action and do not induce cross-resistance to one another.¹⁹ This conforms to what has been previously published on the promising intervention of combination therapy.^{20,21} In specific, Shapira-Furman et al. developed a poly(lactic acid-glycolic acid) (PLGA)-based wafer that continuously released 5 mg TMZ and 0.38 mg BCNU for 35 days in vitro and tested its anticancer efficacy in vivo in an intracranial 9L gliosarcoma model in F344 rats.¹⁵ Animals treated with TMZ/BCNU wafer had the longest median survival of 28 days with 25% long-term survivors (>120) days, whereas animals in the untreated group were all dead by 11 days after implantation. Though shown effective in vivo, the preparation process for this polymer is cumbersome and involves complex procedures, multiple organic

solvents such as dichloromethane and heptane, and a long incubation period for solvent decanting. Additionally, a wafer-based system is less capable of intimate contact with surrounding tissues. In another study, Tyler et al. tested how OncoGel, a ReGel carrying PTX at 0.16 mg, affected the long-term survival of an intracranial 9L gliosarcoma model.²² ReGel is a thermosensitive formulation that can alternate between a low viscosity fluid at low temperatures ranging between 2 and 15 °C and a more viscous gel at 37 °C. Animals that received XRT combined with OncoGel 5 days after tumor injection had a significantly enhanced median survival of 32 days with 12.5% of them surviving longer than 120 days. However, this delivery system also has its own limitations, as previously discussed.

In the present study, we combine the advantages of both systems, thermosensitive hydrogels and solid wafers, creating thermosensitive nanogels carrying TMZ and PTX in the form of pellets. The pellets are easy to transport, store, and handle. TMZ in a liquid form often turns pink when exposed to air or stored in an airtight container at room temperature due to the rapid

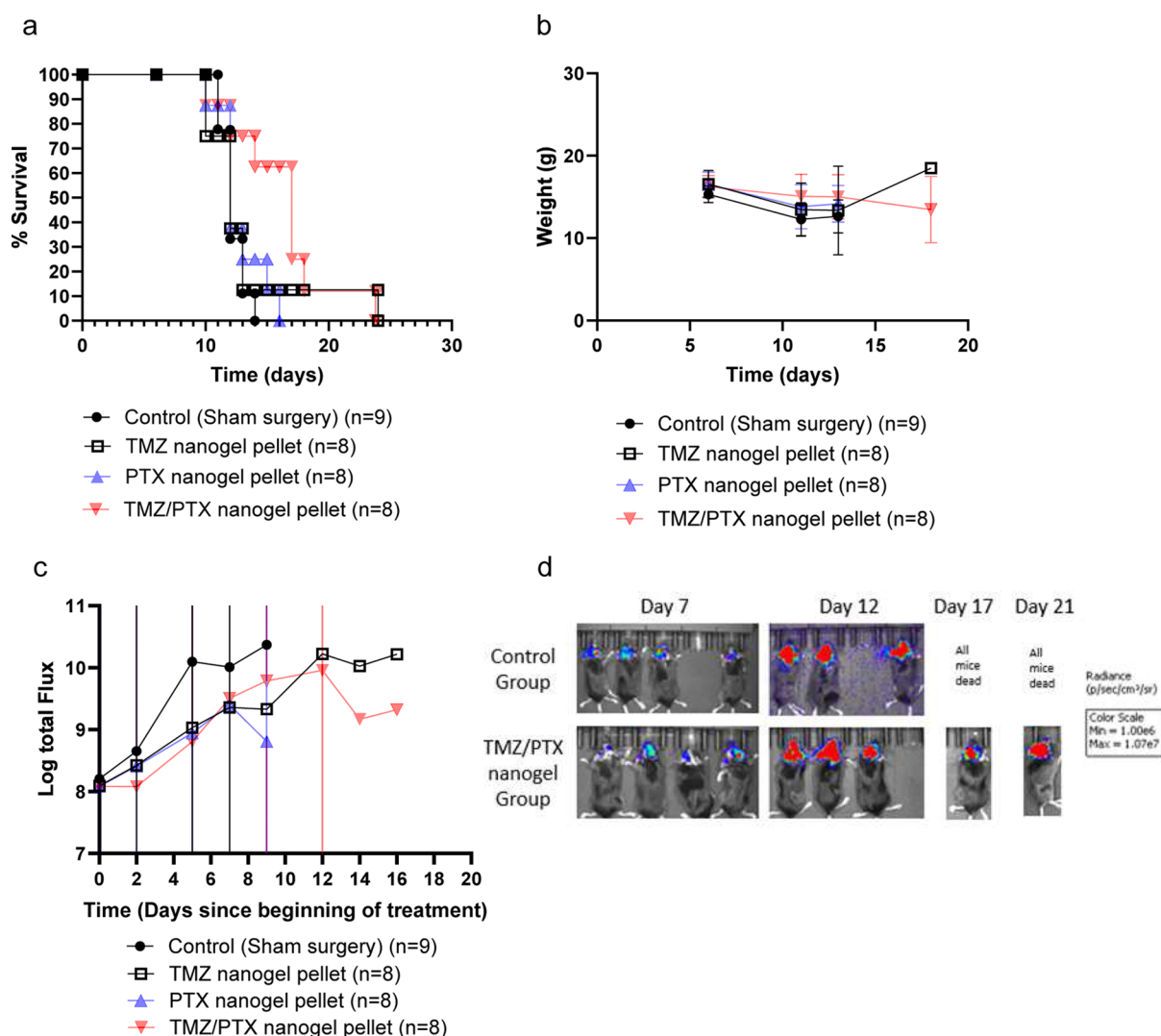


Figure 10. In vivo efficacy study of the drug-loaded nanogel pellets in C57Bl6 mice implanted with murine glioma CT2A cells. Sham surgery indicates that the mice underwent all steps of the surgical procedure that their counterparts in other groups had (anesthesia, incision, stapling) but did not receive any treatment: (a) Kaplan–Meier plot for long-term survival, (b) graph showing the variation of weight measurements of animals over time, (c) graph showing the change in bioluminescence signals (total flux) from CT2A tumor burden over time, and (d) IVIS images of live animals showing bioluminescence from CT2A tumor.

oxidation process. There were no visual changes observed in appearance (e.g., color and shape) for the TMZ and TMZ/PTX pellets when properly stored (e.g., in an airtight container) at room temperature. The pellets carrying PTX and TMZ absorbed small amounts of fluids (less than two-folds of its weight), formed a gel at 37 °C, and released drugs gradually. Due to the relatively more hydrophilic nature of TMZ (log *P* 1.4), the release of TMZ was more rapid than that of PTX (log *P* 3.96) in vitro for both nanogel (sol–gels) and nanogel pellet formulations. The release of the payloads for nanogel pellets was slower than that for nanogels, presumably due to the time required for the pellets to absorb water and form a gel to release the loaded drugs. The drug release was slightly delayed for two drug-loaded formulations compared to the single-drug formulations, as the hydrophobic environment provided by two drugs (TMZ and PTX) held together the nanogels longer, delaying the release of the payloads into the aqueous medium.

The in vivo studies conducted herein using a rat model of brain cancer showed a significant survival advantage with the implanted TMZ/PTX nanogel pellets with and without

radiation in resected models of 9L gliosarcoma. The overall survival achieved by this combination was similar to that achieved by 3.85% BCNU wafers and oral TMZ plus radiation, which are two FDA-approved treatment regimens that are routinely used in the clinical management of glioblastoma.^{13,14,23} Additionally, in both rat studies, the TMZ/PTX nanogel pellet group had a notable percentage of long-term survivors, which were not observed in the control groups or the oral TMZ plus radiation group. The comparable efficacy of the combination nanogel pellets to readily established and approved treatment regimens and its positive impact on safety as well as overall, median, and long-term survival vouches for its clinical translatability. The safety and efficacy of this new combination nanogel pellet were rigorously tested in two different cell lines and animal models. In fact, the 5 day period between the implantation of 9L tumor pieces and the beginning of treatment replicates the clinical scenario of patient presentation and allows for the growth of 9L tumors in an aggressive and invasive manner.²⁴ In a similar fashion, the CT2A mouse glioma model is also known to be an aggressive model that is widely used in

murine glioma studies.²⁵ Additionally, the demonstrated positive impact of the TMZ/PTX nanogel pellets on both overall and median survival in an unresected mouse cohort with confirmed establishment of the tumors via in vivo bioluminescence-based imaging showed that the combination nanogel remained effective even in the setting of high tumor burden. Hence, the TMZ/PTX nanogel pellets show promising potential for clinical translation, especially after further exploration into their in situ pharmacokinetics and utility as an adjuvant therapy that can be incorporated with oral TMZ to achieve higher interstitial concentrations of the drug.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.molpharmaceut.4c00708>.

Nanogel pellets stored in artificial cerebrospinal fluids at 22 °C (MP4)

Nanogel pellets stored in artificial cerebrospinal fluids at 37 °C (MP4)

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Notes

The authors declare the following competing financial interest(s): Hyunah Cho is a consultant of OncoGone, Inc. Henry Brem is a paid consultant to Insightec and chairman of the company's Medical Advisory Board. Insightec is developing

focused ultrasound treatments for brain tumors. This arrangement has been reviewed and approved by the Johns Hopkins University in accordance with its conflict-of-interest policies. Henry Brem receives research funding from NIH, Johns Hopkins University, Khatib Foundation, NICO Myriad Corporation, and philanthropy. He is a consultant for Insightec, Candel Therapeutics, Inc., Catalio Nexus Fund II, LLC, LikeMinds, Inc*, Galen Robotics, Inc.* CraniUS*, and Nurami Medical*. Betty Tyler has research funding from NIH. Ashvattha Therapeutics Inc* has licensed one of her patents and she is a stockholder for Peabody Pharmaceuticals* (*includes equity or options). Ashvattha Therapeutics Inc. has also licensed one of her patents, and she is a stockholder in Peabody Pharmaceuticals (*includes equity or options).

REFERENCES

- (1) Singh, N.; Miner, A.; Hennis, L.; Mittal, S. Mechanisms of Temozolomide Resistance in Glioblastoma - a Comprehensive Review. *Cancer. Drug. Resist.* **2021**, *4*, 17.
- (2) Ostrom, Q. T.; Price, M.; Neff, C.; Cioffi, G.; Waite, K. A.; Kruchko, C.; Barnholtz-Sloan, J. S. CBTRUS Statistical Report: Primary Brain and Other Central Nervous System Tumors Diagnosed in the United States in 2016–2020. *Neuro Oncol.* **2023**, *25* (Supplement 4), iv1–iv99.
- (3) Lee, J. H.; Wee, C. W. Treatment of Adult Gliomas: A Current Update. *Brain Neurorehabil.* **2022**, *15* (3), No. e24.
- (4) Stawicki, B.; Schacher, T.; Cho, H. Nanogels as a Versatile Drug Delivery System for Brain Cancer. *Gels* **2021**, *7* (2), 63.
- (5) Hanif, F.; Muzaffar, K.; Perveen, kahkashan.; Malhi, S.; Simjee, S. Glioblastoma Multiforme: A Review of Its Epidemiology and Pathogenesis through Clinical Presentation and Treatment. *APJCP* **2017**, *18* (1), 3.
- (6) Qu, S.; Qi, S.; Zhang, H.; Li, Z.; Wang, K.; Zhu, T.; Ye, R.; Zhang, W.; Huang, G.; Yi, G. Albumin-Bound Paclitaxel Augment Temozolomide Treatment Sensitivity of Glioblastoma Cells by Disrupting DNA Damage Repair and Promoting Ferroptosis. *J. Exp. Clin. Cancer Res.* **2023**, *42* (1), 285.
- (7) Guan, D.-G.; Chen, H.-M.; Liao, S.-F.; Zhao, T.-Z. Combination of Temozolomide and Taxol Exerts a Synergistic Inhibitory Effect on Taxol-Resistant Glioma Cells via Inhibition of Glucose Metabolism. *Mol. Med. Rep.* **2015**, *12* (5), 7705–7711.
- (8) Perry, J.; Chambers, A.; Spithoff, K.; Laperriere, N. on behalf of the Neuro-Oncology Disease Site Group§ of Cancer Care Ontario's Program in Evidence-Based Care. Gliadel Wafers in the Treatment of Malignant Glioma: A Systematic Review. *Curr. Oncol.* **2007**, *14* (5), 189–194.
- (9) Elstad, N. L.; Fowers, K. D. OncoGel (ReGel/Paclitaxel) — Clinical Applications for a Novel Paclitaxel Delivery System. *Adv. Drug Delivery Rev.* **2009**, *61* (10), 785–794.
- (10) Rahman, S.; Haque, T. N.; Sugandhi, V. V.; Saraswat, A. L.; Xin, X.; Cho, H. Topical Cream Carrying Drug-Loaded Nanogels for Melanoma Treatment. *Pharm. Res.* **2023**, *40* (10), 2291–2301.
- (11) Tamargo, R.; Epstein, J.; Reinhard, C.; Chasin, M.; Brem, H. Brain Biocompatibility of a Biodegradable, Controlled-Release Polymer in Rats. *J. Biomed. Mater. Res.* **1989**, *23* (2), 253–266.
- (12) Fung, L.; Shin, M.; Brem, H.; Saltzman, W. Chemotherapeutic Drugs Released from Polymers: Distribution of 1,3-Bis(2-Chloroethyl)-1-Nitrosourea in the Rat Brain. *Pharm. Res.* **1996**, *13* (5), 671–682.
- (13) Brem, H.; Piantadosi, S.; Burger, P. C.; Walker, M.; Selker, R.; Vick, N. A.; Black, K.; Sisti, M.; Brem, S.; Mohr, G.; Muller, P.; Morawetz, R.; Schold, S. C. Placebo-Controlled Trial of Safety and Efficacy of Intraoperative Controlled Delivery by Biodegradable Polymers of Chemotherapy for Recurrent Gliomas. *Lancet* **1995**, *345* (8956), 1008–1012.
- (14) Westphal, M.; Hilt, D. C.; Bortey, E.; Delavault, P.; Olivares, R.; Warnke, P. C.; Whittle, I. R.; Jääskeläinen, J.; Ram, Z. A Phase 3 Trial of Local Chemotherapy with Biodegradable Carmustine (BCNU) Wafers (Gliadel Wafers) in Patients with Primary Malignant Glioma. *Neuro Oncol.* **2003**, *5*, 79.
- (15) Shapira-Furman, T.; Serra, R.; Gorelick, N.; Doglioli, M.; Tagliaferri, V.; Cecia, A.; Peters, M.; Kumar, A.; Rottenberg, Y.; Langer, R.; Brem, H.; Tyler, B.; Domb, A. J. Biodegradable Wafers Releasing Temozolomide and Carmustine for the Treatment of Brain Cancer. *J. Controlled Release* **2019**, *295*, 93–101.
- (16) Wang, F.; Huang, Q.; Su, H.; Sun, M.; Wang, Z.; Chen, Z.; Zheng, M.; Chakroun, R. W.; Monroe, M. K.; Chen, D.; Wang, Z.; Gorelick, N.; Serra, R.; Wang, H.; Guan, Y.; Suk, J. S.; Tyler, B.; Brem, H.; Hanes, J.; Cui, H. Self-Assembling Paclitaxel-Mediated Stimulation of Tumor-Associated Macrophages for Postoperative Treatment of Glioblastoma. *Proc. Natl. Acad. Sci. U.S.A.* **2023**, *120* (18), No. e2204621120.
- (17) Smith, S. J.; Tyler, B. M.; Gould, T.; Veal, G. J.; Gorelick, N.; Rowlinson, J.; Serra, R.; Ritchie, A.; Berry, P.; Otto, A.; Choi, J.; Skuli, N.; Estevez-Cabrero, M.; Shakesheff, K. M.; Brem, H.; Grundy, R. G.; Rahman, R. Overall Survival in Malignant Glioma Is Significantly Prolonged by Neurosurgical Delivery of Etoposide and Temozolomide from a Thermo-Responsive Biodegradable Paste. *Clin. Cancer Res.* **2019**, *25* (16), 5094–5106.
- (18) Bastiancich, C.; Bozzato, E.; Luyten, U.; Danhier, F.; Bastiat, G.; Pr  at, V. Drug Combination Using an Injectable Nanomedicine Hydrogel for Glioblastoma Treatment. *Int. J. Pharm.* **2019**, *559*, 220–227.
- (19) Tseng, S. H. Characterization of Paclitaxel (Taxol(R)) Sensitivity in Human Glioma- and Medulloblastoma-Derived Cell Lines. *Neuro Oncol.* **1999**, *1* (2), 101–108.
- (20) Vellimana, A. K.; Recinos, V. R.; Hwang, L.; Fowers, K. D.; Li, K. W.; Zhang, Y.; Okonma, S.; Eberhart, C. G.; Brem, H.; Tyler, B. M. Combination of Paclitaxel Thermal Gel Depot with Temozolomide and Radiotherapy Significantly Prolongs Survival in an Experimental Rodent Glioma Model. *J. Neuro Oncol.* **2013**, *111* (3), 229–236.
- (21) Zhao, M.; Bozzato, E.; Joudiou, N.; Ghiassinejad, S.; Danhier, F.; Gallez, B.; Pr  at, V. Codelivery of Paclitaxel and Temozolomide through a Photopolymerizable Hydrogel Prevents Glioblastoma Recurrence after Surgical Resection. *J. Controlled Release* **2019**, *309*, 72–81.
- (22) Tyler, B.; Fowers, K. D.; Li, K. W.; Recinos, V. R.; Caplan, J. M.; Hdeib, A.; Grossman, R.; Basaldella, L.; Bekelis, K.; Pradilla, G.; Legnani, F.; Brem, H. A Thermal Gel Depot for Local Delivery of Paclitaxel to Treat Experimental Brain Tumors in Rats: Laboratory Investigation. *JNS* **2010**, *113* (2), 210–217.
- (23) Stupp, R.; Mason, W. P.; van den Bent, M. J.; Bogdahn, U.; Ludwin, S. K.; Lacombe, D.; Mirimanoff, R. O. Radiotherapy plus Concomitant and Adjuvant Temozolomide for Glioblastoma. *N. Engl. J. Med.* **2005**, *27*, 15–16.
- (24) Serra, R.; Mangraviti, A.; Gorelick, N. L.; Shapira-Furman, T.; Alomari, S.; Cecia, A.; Darjee, N.; Brem, H.; Rottenberg, Y.; Domb, A. J.; Tyler, B. Combined Intracranial Acriflavine, Temozolomide and Radiation Extends Survival in a Rat Glioma Model. *Eur. J. Pharm. Biopharm.* **2022**, *170*, 179–186.
- (25) Slika, H.; Karimov, Z.; Alimonti, P.; Abou-Mrad, T.; De Fazio, E.; Alomari, S.; Tyler, B. Preclinical Models and Technologies in Glioblastoma Research: Evolution, Current State, and Future Avenues. *IJMS* **2023**, *24* (22), 16316.